

Lab Exam Part 2

Customer Churn Prediction

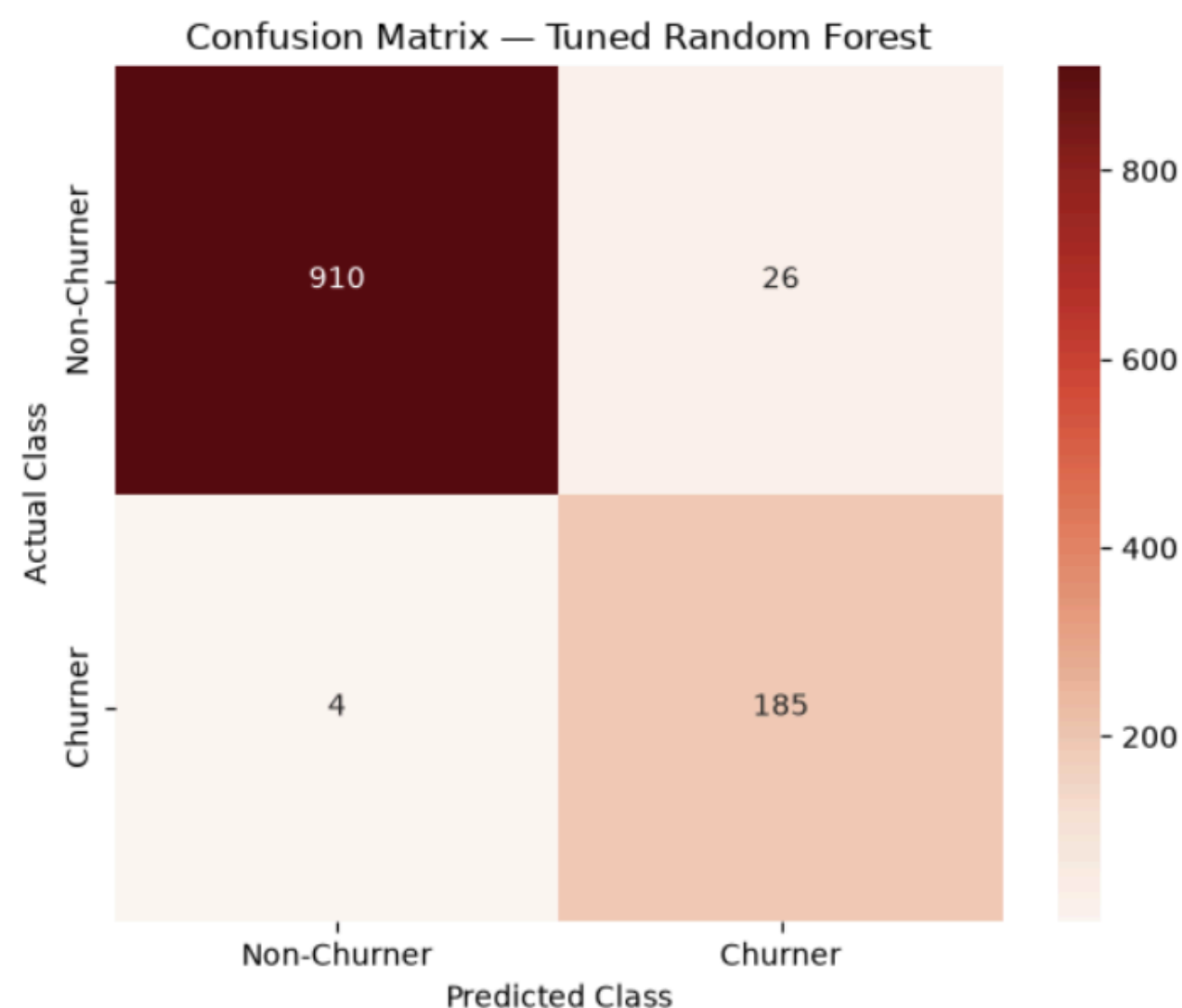
**Propensity Modeling and
Targeted Retention Strategy**

Irene Marrali
AI for Communication and Marketing

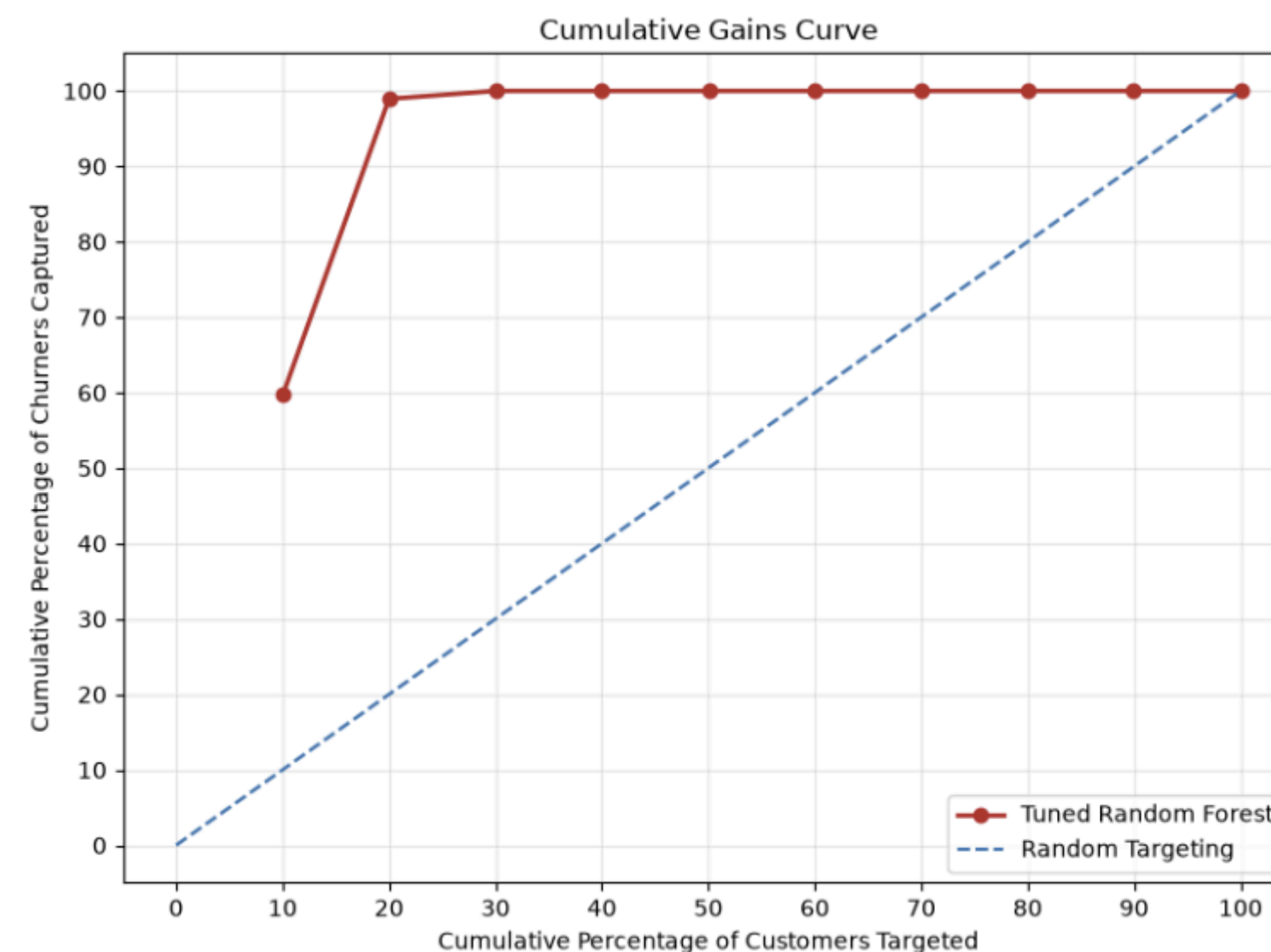


Churn Model Performance

Three models were compared using stratified cross-validation: **Random Forest** achieved the best overall performance, with the highest Recall, and was therefore selected and optimized through Randomized Search. It reached **97.9% recall, 92.5% F1-Score and 99.8% ROC-AUC**.



185 of 189 churners were correctly identified; **only 4 were missed.**



The model captures 98.9% of churners by targeting only **the top 20% of customers.**

What Drives Customers Away?

Feature importance identifies the variables that **contribute most to churn prediction**.

Low Tenure



Customers in their first 3 months show the highest churn rate: 41.8%.

Cashback Behavior

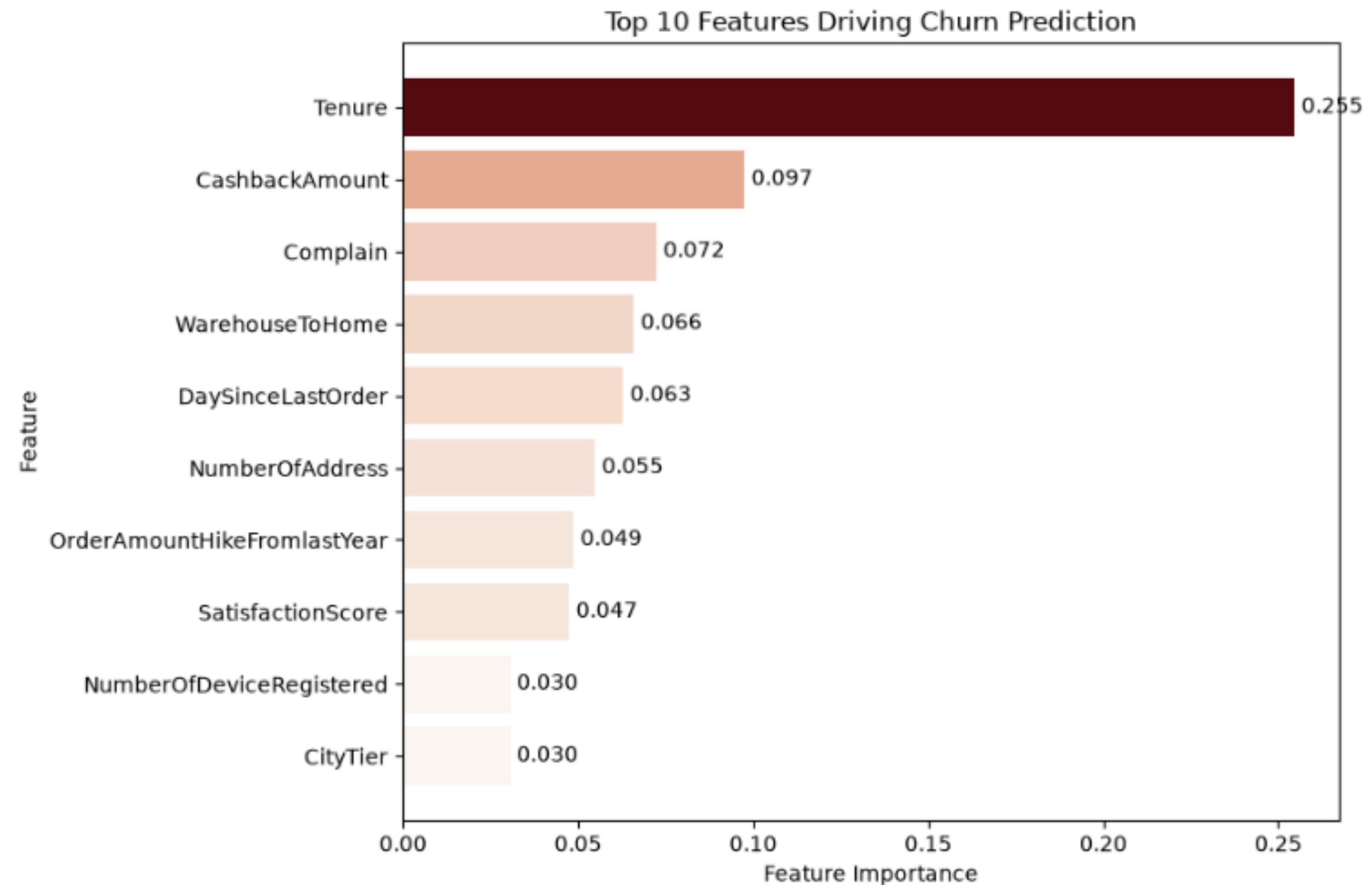


Cashback patterns reveal differences in customer engagement and price sensitivity.

Recent Complaint

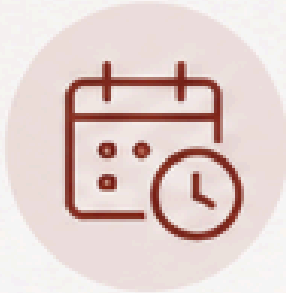


Customers who raised a complaint show a 31.7% churn rate, compared with 10.9% among customers without complaints.



Strategic Recommendation

Personalized retention actions for the top 20% highest-risk customers, based on their main churn driver.

RISK DRIVER	MARKETING STRATEGY	TACTICS & CHANNELS
 LOW TENURE (0–3 months)	 Accelerate early value Help new customers succeed and build habit quickly.	 Welcome series Tips, best practices, product education.  Onboarding incentive Personalized coupon or cashback after key first action.  Milestone nudges Reminders and encouragement to reach next milestones.
 RECENT COMPLAINT (Any time)	 Recover trust quickly Turn a negative experience into a positive one.	 Immediate outreach Personal call or message from customer care.  Personalized resolution Solve the issue and follow up.  Goodwill offer Compensation (e.g., cashback, discount, free delivery).
 CASHBACK BEHAVIOR (Any level)	 Reward & re-engage Increase engagement and purchase frequency in a cost-effective way.	 Personalized cashback Tailored to spend behavior and preferred categories.  Reactivation offer Time-limited incentive for inactive or low-engagement customers.  Loyalty reward Bonus cashback or points for continued activity or higher-value actions.